

Industrial Dual Protocol Tag Simulator Manual

This manual shows how to configure MQTT, OPC UA, and BOTH protocols together with shared or separate CSV/XLSX files.

Quick Start

1	Extract the ZIP to a fresh folder.
2	Open PowerShell in that folder.
3	Run: <code>py -m venv venv</code>
4	Run: <code>.\venv\Scripts\Activate.ps1</code>
5	Run: <code>python -m pip install -r requirements.txt</code>
6	Run: <code>python run_desktop.py</code>
7	Open: <code>http://127.0.0.1:8000</code>

MQTT Basic Configuration

Field	Recommended Value
Protocol	MQTT only, or OPC UA + MQTT together
Host	localhost
Port	1883
Topic Prefix	industrial-tag-simulator
Client ID	industrial-tag-simulator
Username / Password	Blank unless your broker requires login
QoS	0
Retain	Unchecked

1. Generate Data

Use the Generate Data tab to create sample industrial CSV files. Click Generate CSV to create a file. Click Generate and Load to also open it in manual replay mapping.

INDUSTRIAL DUAL PROTOCOL TAG SIMULATOR
Manual screenshot - Generate CSV/XLSX data

1. GENERATE DATA

2. FILES / ASSIGNMENT

3. REPLAY OPC UA / MQTT

4. CONFIGS / STATUS

1

Domain + Scenario
Choose generator, for example EAF Melting, Pipeline, Power Plant

2

Parameters
Set row count, start time, leak/scenario options, frequency, etc.

3

Generate CSV
Creates a new CSV in generated_data.

4

Generate and Load
Creates CSV and opens it in manual tag mapping.

2. Upload and Assign Files

Use Files / Assignment to upload CSV/XLSX files and assign them to Shared, OPC UA, or MQTT. For different files per protocol, select OPC UA for one file and MQTT for another file.

INDUSTRIAL DUAL PROTOCOL TAG SIMULATOR
Manual screenshot - Assign files

1. GENERATE DATA

2. FILES / ASSIGNMENT

3. REPLAY OPC UA / MQTT

4. CONFIGS / STATUS

1

Upload CSV/XLSX
Upload one or more files. Existing sample/generated files also appear here.

2

Available Files
Use checkboxes per file: Shared, OPC UA, MQTT. You can assign different files to each protocol.

3

Preview
Click Preview to check columns and rows before running.

4

Current Assignment
Confirm which files are Shared, OPC UA-only, and MQTT-only.

3. Configure MQTT

Go to Replay OPC UA / MQTT. Select MQTT only or OPC UA + MQTT together. Enter your MQTT broker details, then click Apply Selected File Assignment and Start.

INDUSTRIAL DUAL PROTOCOL TAG SIMULATOR
Manual screenshot - Configure MQTT

1. GENERATE DATA

2. FILES / ASSIGNMENT

3. REPLAY OPC UA / MQTT

4. CONFIGS / STATUS

1

Protocol Mode
Select MQTT only or OPC UA + MQTT together.

2

MQTT Settings
Host: localhost
Port: 1883
Topic Prefix: industrial-tag-simulator
Client ID: industrial-tag-simulator
QoS: 0, Retain unchecked

3

Apply Selected File Assignment
Uses the file checkboxes from Files / Assignment.

4

Start / Stop
Start replay after applying config. Keep terminal open.

Test with Mosquitto subscriber:

```
mosquitto_sub -h localhost -p 1883 -t "industrial-tag-simulator/#" -v
```

4. Configure OPC UA and BOTH Mode

For OPC UA, keep the root folder as TagSimulator to avoid duplicate roots. For BOTH mode, assign files separately in Files / Assignment, then choose OPC UA + MQTT together.

INDUSTRIAL DUAL PROTOCOL TAG SIMULATOR
Manual screenshot - Run OPC UA and MQTT together

1. GENERATE DATA

2. FILES / ASSIGNMENT

3. REPLAY OPC UA / MQTT

4. CONFIGS / STATUS

1

Protocol Mode
Choose OPC UA only, MQTT only, or BOTH together.

2

OPC UA Settings
Namespace URI, Root Folder, and Node ID Prefix. Root Folder should stay TagSimulator.

3

Both Protocols With Different Files
Assign OPC UA files and MQTT files separately in Files / Assignment, then choose BOTH here.

4

Runtime Status
OPC UA endpoint and MQTT status are shown at the top and in Configs / Status.

OPC UA endpoint:

```
opc.tcp://localhost:4840/simulator
```

Common Workflows

MQTT only: Assign file to MQTT, choose MQTT only, apply selected file assignment, start.

OPC UA only: Assign file to OPC UA, choose OPC UA only, apply selected file assignment, start.

BOTH same file: Assign file to Shared, choose OPC UA + MQTT together, apply selected file assignment, start.

BOTH different files: Assign 1.csv to OPC UA and 2.csv to MQTT, choose OPC UA + MQTT together, apply selected file assignment, start.

Fix in this build

The UI now safely initializes protocol status before reading OPC UA/MQTT running state. The error Cannot read properties of undefined (reading running) is prevented, and browser cache is bypassed using a new app.js version query.